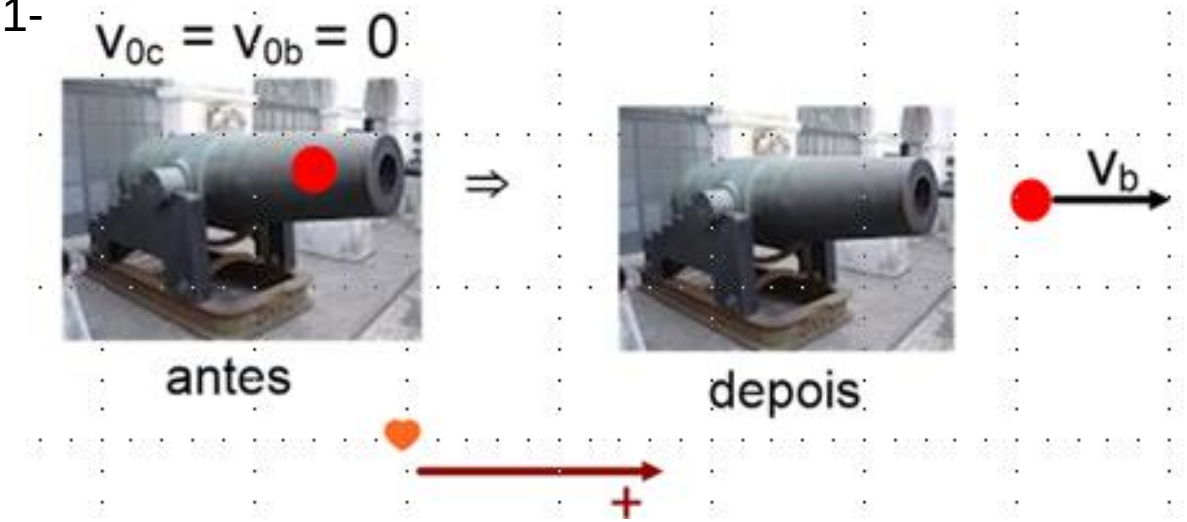
1-



b) $Q_{antes} = Q_{depois}$

$M.v_{oc} + m.v_{ob} = M.v_c + m.v_b$

$0 = 300.v_c + 15.60$

$v_c = -3\ m/s$

c) $v_{c,b} = v_c - v_b$

$v_{c,b} = -3 - (60)$

$v_{c,b} = -63\ m/s$

d)

$$E_{cantes} = \frac{m_c \, v^2_{0c}}{2} + \frac{m_b \, v^2_{0b}}{2} \Rightarrow E_{cantes} = 0$$

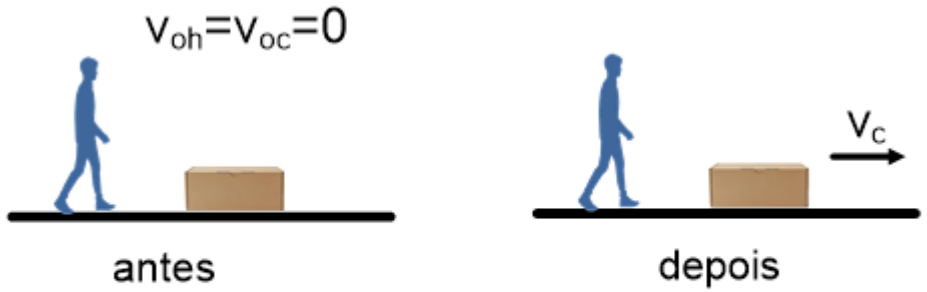
$$*E_{cdepois} = \frac{m_c \, v_c^2}{2} + \frac{m_b \, v_b^2}{2}$$

$$* E_{cdepois} = \frac{300^{(-3)^2}}{2} + \frac{15.60^2}{2} = 28350 \text{ J}$$

$$* \ \Delta E_c = E_{cdepois} - E_{cantes}$$

$$\Delta E_c = 28350 \text{ J}$$

2-



$$Q_{antes} = Q_{depois}$$

$$m_h.v_{0h}+m_c.v_{0c} = m_h.v_{oh}+m_c.v_{oc}$$

$$0= 80.v_h + 40.10$$

$$v_h = -5\ m/s$$

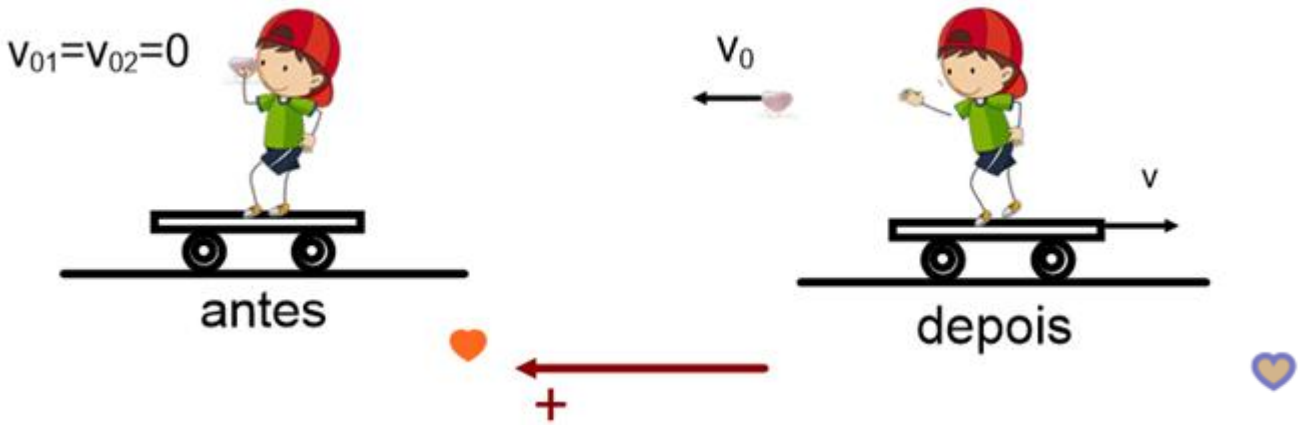
3-

*1 = pedra

$m_1 = 2$ kg e $v_0 = 3$ m/s

*2 = garoto + carrinho

$m_2 = m_g + m_c$ --- $m_2 = 50 + 10 = 60$ kg



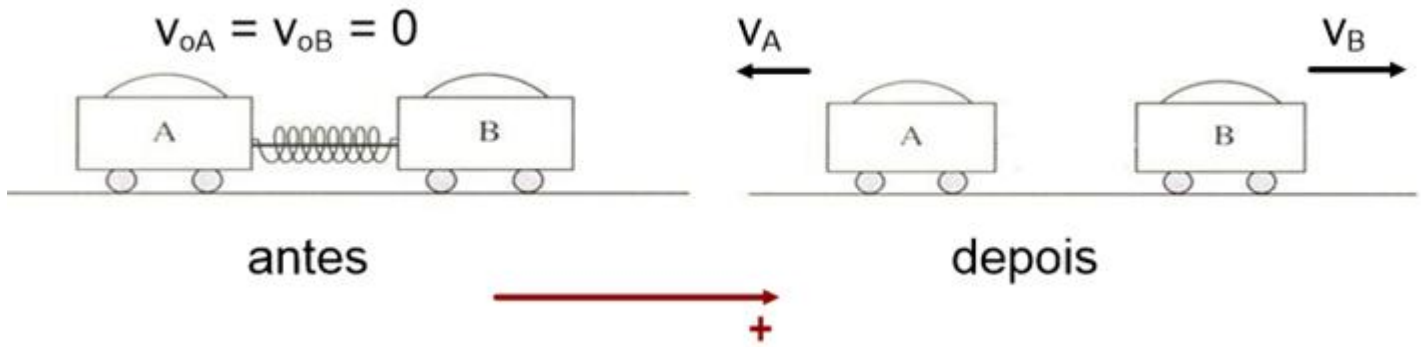
$Q_{antes} = Q_{depois}$

$m_1.v_{01} + m_2.v_{02} = m_1.v_1 + m_2.v_2$

$0 = 2.3 + 60.v$

$v = -0,1$ m/s

4-



$$Q_{antes} = Q_{depois}$$

$$m_A.v_{oA} + m_B.v_{oB} = m_A.v_A + m_B.v_B$$

$$0 = 4.(-3)+2.v_B$$

$$v_B = 6\ m/s$$

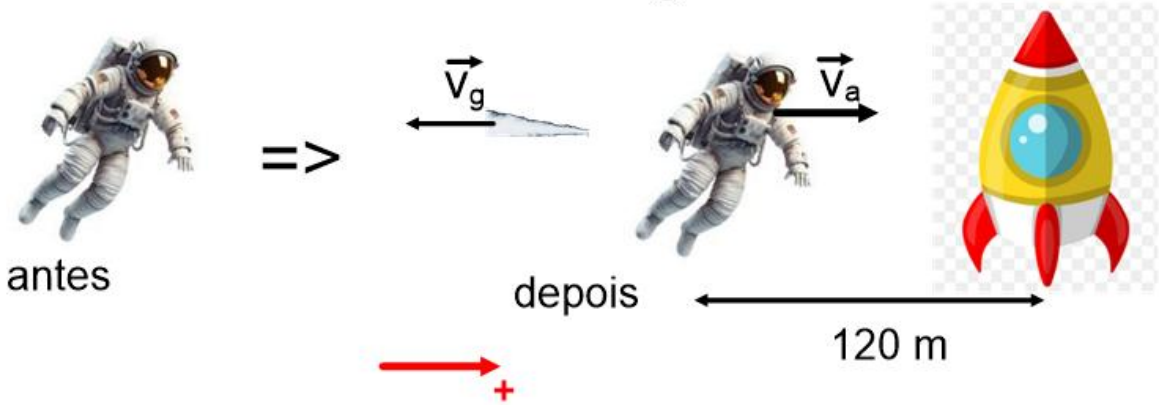
$$E_{Mantes} = E_{Mdepois}$$

$$E_{pe} = E_{cA} + E_{cB}$$

$$E_{pe} = \frac{4.9}{2} + \frac{2.36}{2} = 54\ J$$

5-

Sistema: Astronauta + gás



$$Q_{0sist} = Q_{sist}$$

$$(m_a+m_g).v_0 = m_a.v_a + m_g.v_g$$

$$0 = 70.v_a + 0,1.(-5,6.10^2)$$

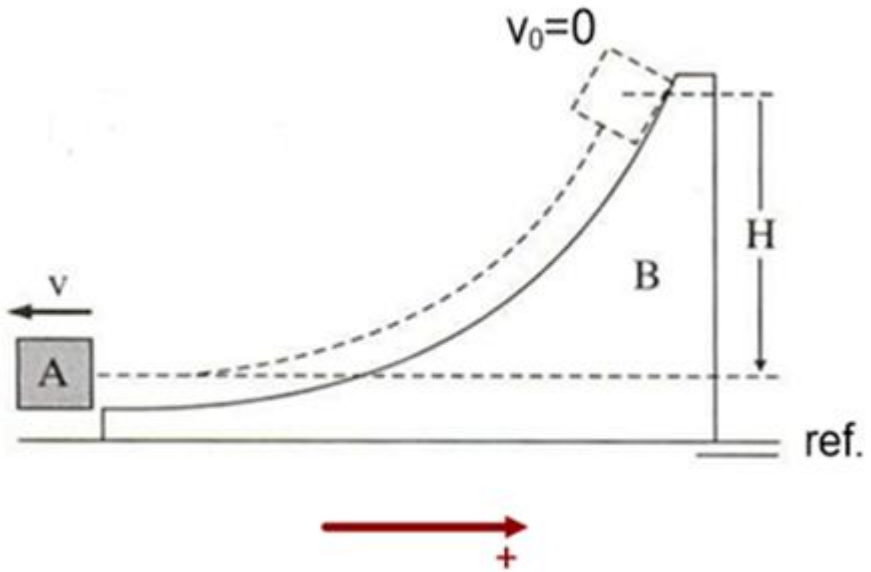
$$70.v_a = 56$$

$$v_a = 0,8 \text{ m/s}$$

$$v = \frac{\Delta s}{\Delta t}$$

$$0,8 = \frac{120}{\Delta t} \quad => \quad \Delta t = 2 \text{ min } 30 \text{ s}$$

6-



$m_B = M$ $v_0 = 0$

$m_A = M/2$ $v = 2$ m/s

★ a) $Q_{antes} = Q_{depois}$

$$(m_A + m_B).v_o = m_A.v_A + m_B.v_B$$

$$0 = \frac{M.(-2)}{2} + M.v_B$$

$$v_B = 1 \text{ m/s}$$

b) $E_{Mantes} = E_{Mdepois}$

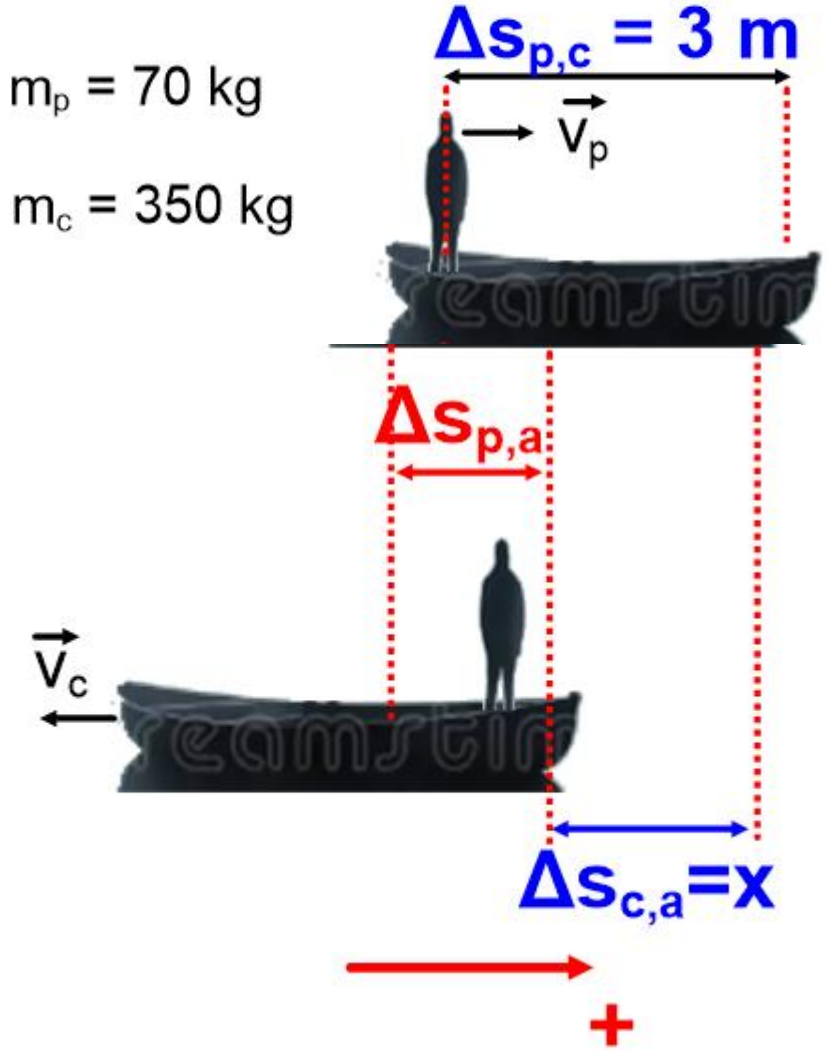
$$Ep_A = Ec_A + Ec_B$$

$$\frac{M}{2}.g.H = \frac{M}{2}.\frac{v_A^{\,2}}{2} + \frac{M.v_B^{\,2}}{2}$$

$$10.H = \frac{4}{2} + 1$$

$$H = 0{,}3 \text{ m}$$

7-



$\Delta s_{p,c} = 3m$

$\Delta s_{c,a} = x$

$\Delta s_{p,a} = 3 - x$

$$Q_{0sist} = Q_{sist}$$

$$m_p.v_{op} + m_c.v_{0c} = m_p.v_p + m_c.v_c$$

$$0 = 350.\frac{\Delta s_c}{\Delta t} + 70.\frac{\Delta s_p}{\Delta t}$$

$$0 = 350.(-x) + 70.(3-x)$$

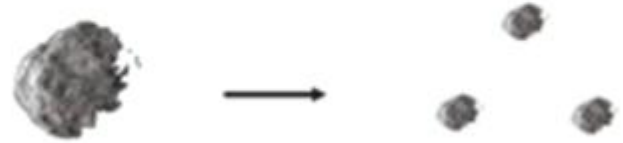
$$0 = 5.(-x) + 3 - x$$

$$6.x = 3 \Rightarrow x = 0,5\ m$$

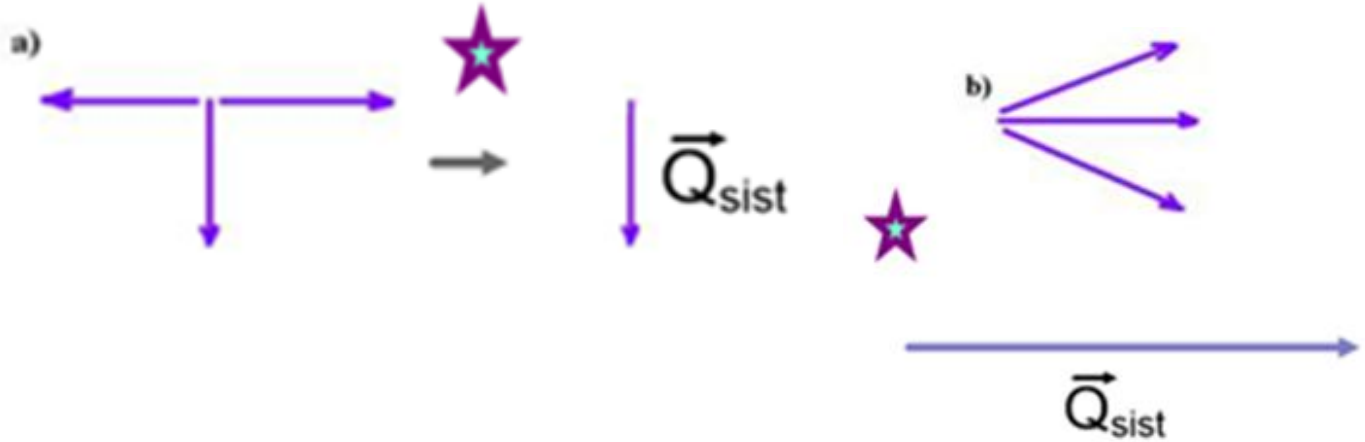
8-

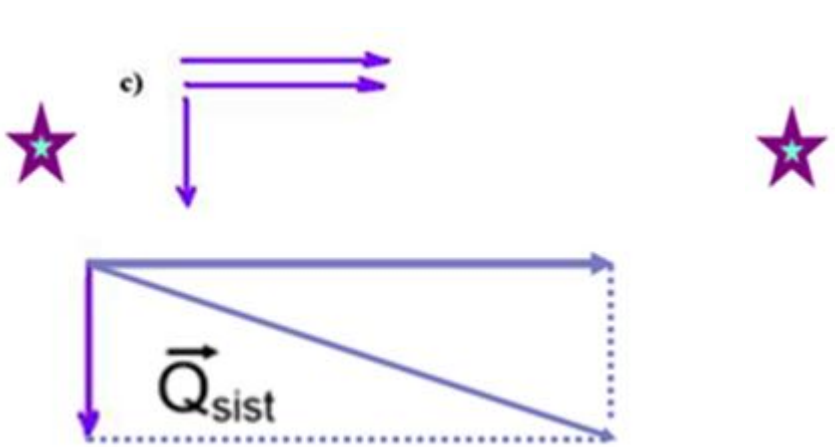
$v_0 = 0$ $\quad \vec{Q}_{sist} = \vec{Q}_{0sist}$

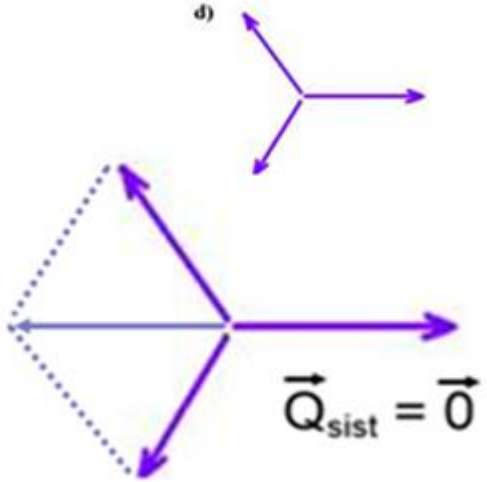
$\vec{Q}_{0sist} = \vec{0}$ $\quad \vec{Q}_{sist} = \vec{0}$

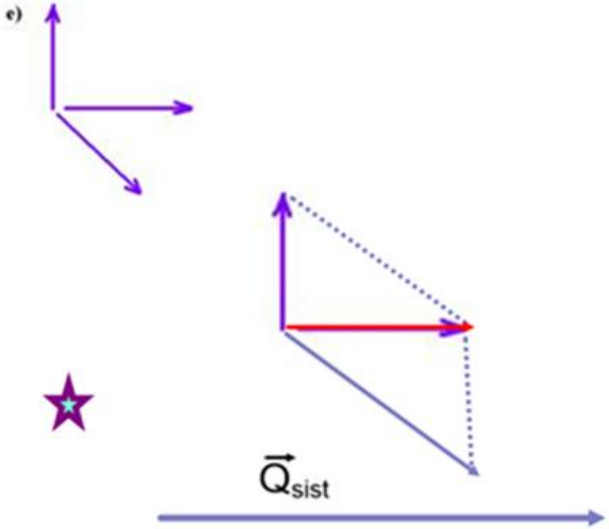


antes depois



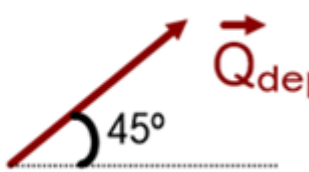




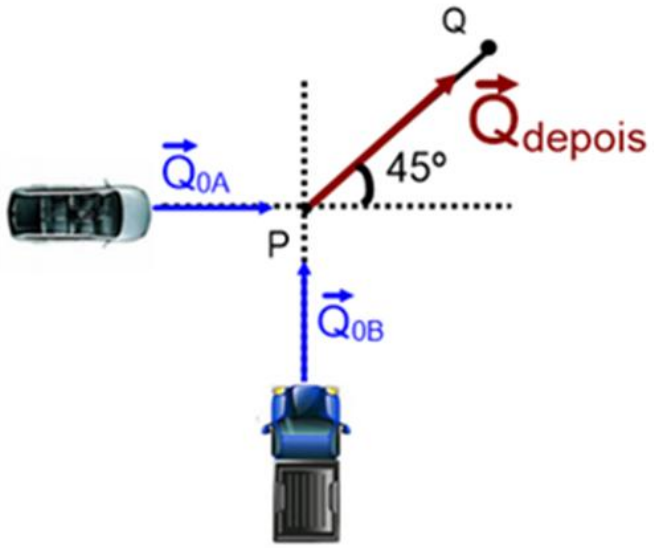


9-

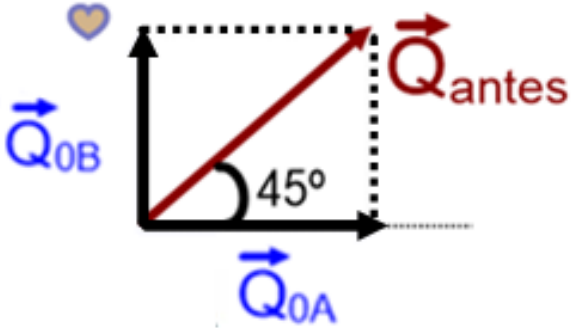
$m_A = m$; $m_B = 3m$



$\vec{Q}_{depois} = \vec{Q}_{antes}$



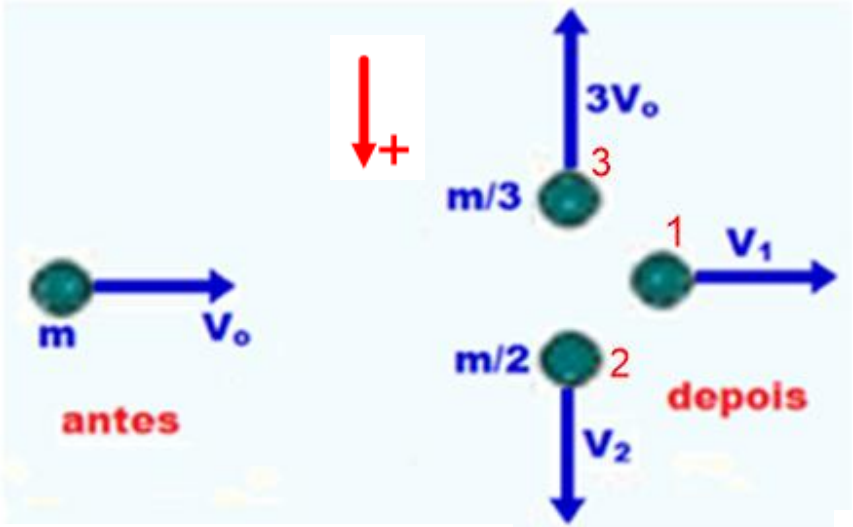
$\vec{Q}_{antes} = \vec{Q}_{0A} + \vec{Q}_{0B}$



$$tg\ 45° = \frac{Q_{0B}}{Q_{0A}}$$

$$tg\ 45° = \frac{m_B.v_{0B}}{m_A.v_{0A}} \Rightarrow 1 = \frac{3m.36}{m.v_{0A}} \Rightarrow v_{0A} = 108\ km/h$$

10-



*Vertical

$Q_{antes} = Q_{depois}$

$0 = \frac{m}{3}.(-3v_0) + \frac{m}{2}.v_2$

$v_2 = 2v_0$

*Horizontal

$Q_{antes} = Q_{depois}$

$m.v_0 = \frac{m}{6}.v_1$

$v_1 = 6v_0$

$m = \frac{m}{3} + \frac{m}{2} + m_1$

$m - \frac{m}{3} - \frac{m}{2} = m_1$

$\frac{6m-2m-3m}{6} = m_1$

$m_1 = \frac{m}{6}$